Volume of a d-Simplex in Terms of its $(d-2)$-Faces

Rory Shaughnessy

Supervisor: Dr. James Cruickshank

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1 Introduction

Simplices and their properties have a wide range of uses in applications. What fascinates me, however, is not those applications but the simplices themselves as objects of study. There are still open questions in geometry, even in low dimensions, where one might expect nothing of interest to remain.

In this project we explore a viewpoint that moves away from edge lengths and asks whether the volume of a d-simplex can be expressed in terms of another natural family of quantities: the volumes of its $(d-2)$-dimensional faces. At first this choice seems arbitrary, but there are exactly as many $(d-2)$-faces as edges, so one might hope that they carry the same information. We quickly find how subtle the problem is. Along the way we study related questions about recovering the volume of a tetrahedron from partial information.

My personal goal for this month-long summer project was to get a sense of how mathematical research is carried out: to read about an open topic, study it, and see what progress I could make in a month, even as a penultimate-year student. I had produced projects for modules before, but they lacked the originality of this internship; that originality was both a difficulty and a great source of motivation.

2 Background

2.1 A d-simplex

Definition 2.1. A d-simplex is the convex hull of $d+1$ affinely independent points $p_0, \dots, p_d \in \mathbb{R}^d$. We write $[p_0\, p_1 \cdots p_d]$ for it and Δ^d for the collection of all d-simplices in $\mathbb{R}^d$.

For example, a 1-simplex is a line segment, a 2-simplex is a triangle and a 3-simplex is a tetrahedron. Equivalently:

Definition 2.2. For affinely independent $p_0, \dots, p_d \in \mathbb{R}^d$,

$$[p_0 \cdots p_d] = \left\{ \sum_{i=0}^{d} a_i p_i \;\middle|\; \sum_{i=0}^{d} a_i = 1,\ a_i \geq 0 \text{ for } i = 0, \dots, d \right\}.$$

A 0-simplex is a single point.

The two definitions agree, since the convex hull of a finite set is exactly the set of its convex combinations.

2.2 The Cayley–Menger determinant

Throughout, we let

$$l_{ij} = \|p_i - p_j\|^2$$

denote the *squared* distance between p_i and p_j. The Cayley–Menger (CM) formula gives the volume V of $[p_0 \cdots p_d]$ as

$$V^2 = \frac{(-1)^{d+1}}{(d!)^2\, 2^d}\, D, \qquad D = \det \begin{pmatrix} 0 & 1 & 1 & 1 & \cdots & 1 \\ 1 & 0 & l_{01} & l_{02} & \cdots & l_{0d} \\ 1 & l_{01} & 0 & l_{12} & \cdots & l_{1d} \\ \vdots & \vdots & & & \ddots & \vdots \\ 1 & l_{0d} & l_{1d} & l_{2d} & \cdots & 0 \end{pmatrix}.$$

The matrix is $(d+2) \times (d+2)$ and $D \in \mathbb{Z}[l_{01}, \dots, l_{d-1,d}]$ is homogeneous of degree d in the l_{ij}. The formula generalises Heron's formula for the area of a triangle, and computes the content of any $T \in \Delta^d$ from its $\binom{d+1}{2}$ edge lengths. In particular, the edge lengths determine T up to congruence.

2.3 Faces of simplices

Each face of a simplex is again a simplex, of lower dimension: the convex hull of any $k+1$ of the vertices is a k-simplex, called a *k-face*. (The boundary of a d-simplex is the union of its $d+1$ facets, which are $(d-1)$-simplices.) For a tetrahedron,

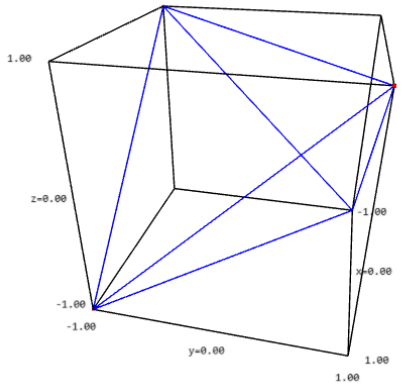


Figure 1: $T \in \Delta^3$ embedded in $\mathbb{R}^3$

we can identify 4 vertices, 6 edges, 4 triangular faces and 1 tetrahedron; that is, 4 0-faces, 6 1-faces, 4 2-faces and one 3-face. In general a d-simplex has $\binom{d+1}{k+1}$ k-faces.

The CM formula expresses the volume of a d-simplex in terms of its 1-faces. Our goal is to investigate the analogous question for its $(d-2)$-*faces*.

For $d = 3$ the $(d-2)$-faces are the edges, so the question is answered by the CM formula. We therefore use $d = 4$ as our running example: it is the first case where the question is non-trivial.

- We look for a formula, or polynomial relation, for the volume of a d-simplex in terms of the $(d-2)$-dimensional volumes of its $(d-2)$-faces, if one exists.
- Ideally such a relation would be a "dual" of the CM formula and share its key features, such as its symmetry under relabelling of the vertices.

2.4 Symmetries of a regular d-simplex

Let $R^d = [p_0 \cdots p_d]$ be a regular d-simplex, i.e. all edges have equal length. Every permutation σ of the vertices is realised by an isometry of $\mathbb{R}^d$: the

affine map sending $p_i \mapsto p_{\sigma(i)}$ preserves all pairwise distances between vertices, and an affine map that preserves the distances between $d+1$ affinely independent points is an isometry. Conversely, an isometry mapping R^d to itself permutes its vertices and is determined by that permutation, since an affine map is determined by its values on $d+1$ affinely independent points. Hence the symmetry group satisfies

$$G_{R^d} \cong S_{d+1}.$$

For R^4 we obtain $|G_{R^4}| = 5! = 120$ symmetries.

2.5 Symmetries of a general d-simplex

Definition 2.3. A map $f\colon \mathbb{R}^d \to \mathbb{R}^d$ is an *isometry* if $\|a-b\| = \|f(a)-f(b)\|$ for all $a, b \in \mathbb{R}^d$. Every isometry is a composition of a translation with an orthogonal linear map (rotations and reflections).

Two simplices $T, Q \in \Delta^d$ are *congruent*, $T \cong Q$, if some isometry maps one onto the other; equivalently, after relabelling vertices they have the same squared edge lengths l_{ij}.

For a general $T \in \Delta^d$, let $\mathrm{Sym}(T)$ be the group of isometries of $\mathbb{R}^d$ mapping T to itself. By the argument above, each element permutes the vertices and is determined by that permutation, so we get an injective homomorphism

$$\mathrm{Sym}(T) \hookrightarrow S_{d+1}.$$

A permutation σ lies in the image exactly when $l_{\sigma(i)\sigma(j)} = l_{ij}$ for all $i < j$. For a generic simplex (all edge lengths distinct) the group is trivial. The regular simplex is the extreme case with the full group S_{d+1}.

Independently of the geometry, S_{d+1} acts on the labels, and hence on the edge set and the face set below. Any relation between edge data and volume, such as the CM formula, is invariant under this action.

2.6 Notation

We let the set of 1-faces (edges) be indexed by

$$E = \{\{i,j\} \,:\, 0 \le i < j \le d\}, \qquad |E| = \binom{d+1}{2} =: k.$$

Similarly, for $I = \{i_0 < \cdots < i_{d-2}\} \subset \{0, \ldots, d\}$ with $|I| = d-1$, let f_I denote the $(d-2)$-dimensional volume of the $(d-2)$-face spanned by $\{p_i : i \in I\}$. We call

$$F = (f_I)_{|I|=d-1}$$

the *face-set* of the simplex. There are $\binom{d+1}{d-1} = \binom{d+1}{2} = k$ such faces, so $|F| = |E|$.

This equality is not just the symmetry of Pascal's triangle. The map

$$\{i, j\} \longmapsto \{0, \ldots, d\} \setminus \{i, j\}$$

is a bijection from edges to $(d-2)$-faces that commutes with the action of S_{d+1}: each $(d-2)$-face is paired with the unique edge disjoint from it, its "opposite" edge.

Each f_I is itself determined by the edges: applying the CM formula to the face, f_I^2 is a polynomial of degree $d-2$ in the l_{ij} with $i, j \in I$. This defines the polynomial map

$$\Phi \colon \mathbb{R}^k \to \mathbb{R}^k, \qquad (l_{ij}) \longmapsto (f_I^2).$$

Definition 2.4. Write $V(T)$ for the volume of $T \in \Delta^d$. By the CM formula, V^2 is a polynomial in the l_{ij}.

The problem can now be stated precisely.

Question 2.5. Is $V(T)$ determined by the face-set F of T? More strongly, is there a function, or even a polynomial $p \in \mathbb{R}[y_I]$, with $V^2 = p\big((f_I^2)\big)$ for every $T \in \Delta^d$?

2.6.1 Example: $d = 4$

For $T \in \Delta^4$, the ten squared edge lengths are

$$l_{01}, l_{02}, l_{03}, l_{04}, l_{12}, l_{13}, l_{14}, l_{23}, l_{24}, l_{34},$$

and the face-set consists of the ten triangle areas

$$F = (f_{012}, f_{013}, f_{014}, f_{023}, f_{024}, f_{034}, f_{123}, f_{124}, f_{134}, f_{234}).$$

Each f_{ijk}^2 is a quadratic polynomial in l_{ij}, l_{ik}, l_{jk} (Heron's formula), and V^2 is a quartic in all ten l_{ij}.

The original plan was to compute directly: eliminate the l_{ij} from the eleven equations

$$y_I = f_I^2(l), \qquad v = V^2(l)$$

with SAGE (a Gröbner basis computation) and study the resulting relation between v and the y_I, using its symmetry and homogeneity. In practice this elimination did not finish in the time available, so the analysis had to be rephrased.

2.7 Algebraic independence and finiteness

The equality $|E| = |F|$ suggests that the face-set carries "the same amount of information" as the edges. The right language for this is algebraic independence.

Definition 2.6. Polynomials $g_1, \dots, g_r \in \mathbb{F}[x_1, \dots, x_n]$ are *algebraically independent* over $\mathbb{F}$ if there is no non-zero $q \in \mathbb{F}[y_1, \dots, y_r]$ with $q(g_1, \dots, g_r) = 0$. Otherwise they are *algebraically dependent*.

Lemma 2.7. *Any $n+1$ polynomials $g_1, \dots, g_{n+1} \in \mathbb{F}[x_1, \dots, x_n]$ are algebraically dependent.*

Proof. Let $D = \max_i \deg g_i$. For $N \geq 1$, consider the products $g^\alpha = g_1^{\alpha_1} \cdots g_{n+1}^{\alpha_{n+1}}$ with $|\alpha| \leq N$. There are $\binom{N+n+1}{n+1}$ of them, which grows like $N^{n+1}/(n+1)!$. Each has degree at most DN, so they all lie in the space of polynomials of degree at most DN in n variables, which has dimension $\binom{DN+n}{n}$; this grows like $(DN)^n/n!$. For N large enough, the number of products exceeds this dimension, so there is a non-trivial linear relation $\sum_\alpha c_\alpha g^\alpha = 0$. Then $q = \sum_\alpha c_\alpha y^\alpha \neq 0$ satisfies $q(g_1, \dots, g_{n+1}) = 0$. $\square$

Applied to the $k+1$ polynomials f_I^2 and V^2 in the k variables l_{ij}, Lemma 2.7 gives a non-zero polynomial q with

$$q\big((f_I^2),\, V^2\big) = 0.$$

On its own this says nothing about V: the relation might not involve V at all. The key input is the following theorem.

Theorem 2.8 (Mohar–Rivin [1])**.** *For $d \geq 3$, the $k = \binom{d+1}{2}$ polynomials f_I^2 are algebraically independent. Moreover, the Jacobian of Φ is non-singular at the regular simplex.*

Their proof computes the Jacobian at the regular simplex. It is a non-zero multiple of the edge–face incidence matrix, whose eigenvalues they compute using Kneser graphs.

Corollary 2.9. *There is a polynomial $q(y, v) = \sum_{j=0}^{m} c_j(y)\, v^j$ with $m \geq 1$ and $c_m \neq 0$ such that $q\big((f_I^2), V^2\big) = 0$ on all of Δ^d. Consequently, for every T outside the proper algebraic subset $\{c_m(\Phi(l)) = 0\}$, the simplices with the same face-set as T take at most m distinct volumes.*

Proof. Take $q \neq 0$ from Lemma 2.7. If q did not involve v, it would be a non-trivial relation among the f_I^2, contradicting Theorem 2.8. So $m \geq 1$, and we may take $c_m \neq 0$. Again by Theorem 2.8, $c_m\big((f_I^2(l))\big)$ is a non-zero polynomial in the l_{ij}, so it vanishes only on a proper algebraic subset. If T' has the same face-set as T and $c_m(\Phi(T)) \neq 0$, then $V(T')^2$ is a root of the non-zero polynomial $v \mapsto q(\Phi(T), v)$ of degree m, which has at most m roots. □

So, generically, the face-set determines the volume up to finitely many possibilities, i.e. V^2 is a *multivalued algebraic function* of the f_I^2. Gaifullin [3] studies this function in detail for 2-faces. Whether the face-set determines V *uniquely* is a separate question; we answer it in Section 4.4.

3 Volume of a 3-simplex from faces and two edges

For a tetrahedron the volume is determined by its six edges. As a warm-up for working with face data, we ask what can be recovered from the four face areas together with two edges that share a vertex. That is $4 + 2 = 6$ pieces of data, matching the six edges.

3.1 Setup

Let $[A, B, C, D]$ be a tetrahedron with edges

$$a = |BC|, \quad b = |CA|, \quad c = |AB|, \quad d = |DA|, \quad e = |DC|, \quad f = |DB|,$$

and face areas $f_{ABC}, f_{ACD}, f_{BCD}, f_{ABD}$. We are given the four face areas and the two edges a, b, which meet at C.

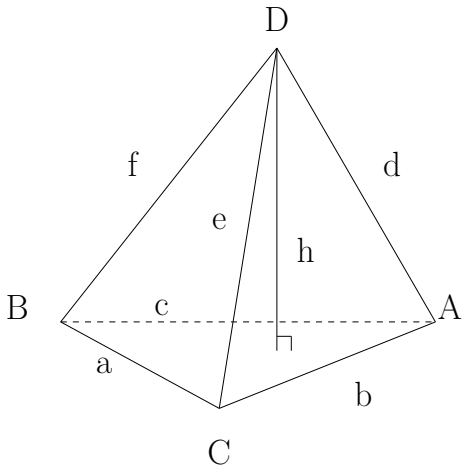


Figure 2: One possible tetrahedron with the given data.

Even the base triangle ABC is not determined. From $f_{ABC} = \frac{1}{2}ab\sin\gamma$, where $\gamma = \angle BCA$, we get $\sin\gamma = 2f_{ABC}/(ab)$. This requires $2f_{ABC} \leq ab$ and leaves two choices, γ and $\pi - \gamma$, which coincide only when $\gamma = \pi/2$. The two choices give $c^2 = a^2 + b^2 \mp 2ab\cos\gamma$.

So uniqueness up to isometry already fails at the base. We now count how many tetrahedra, and how many volumes, are possible.

3.2 Solution

Fix one of the two base triangles. Using isometries of $\mathbb{R}^3$ we may assume:

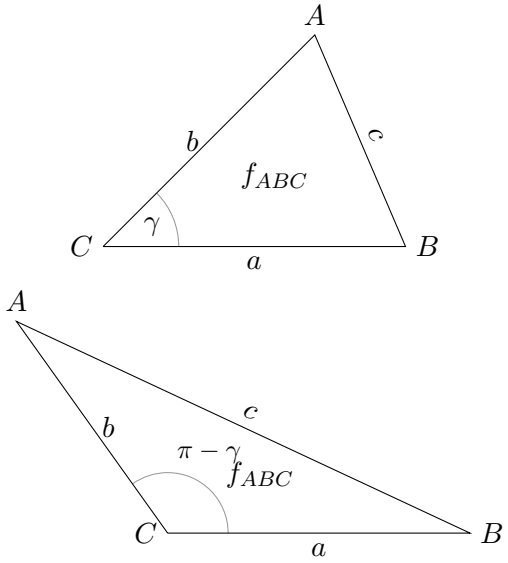


Figure 3: The two possible base triangles ABC with area f_{ABC} (not to scale).

- $C = (0, 0, 0)$ (a translation);
- $B = (a, 0, 0)$ (a rotation);
- $A = (b\cos\gamma,\, b\sin\gamma,\, 0)$ with $b\sin\gamma > 0$ (a reflection in the xz-plane if necessary);
- $D = (x, y, z)$ with $z \geq 0$ (a reflection in the xy-plane if necessary).

After these normalisations, the only remaining isometry preserving the base is the reflection $z \mapsto -z$.

The three side faces are shown below. Their heights from D are

$$h_a = \frac{2f_{BCD}}{a}, \qquad h_b = \frac{2f_{ACD}}{b}, \qquad h_c = \frac{2f_{ABD}}{c}.$$

Thus D lies at distance h_a from the line CB, at distance h_b from the line CA, and at distance h_c from the line AB. In other words, D lies on the intersection of three cylinders C_a, C_b, C_c, with axes the three sides of the base triangle and radii h_a, h_b, h_c.

Write $P = (x, y)$ for the orthogonal projection of D to the base plane, and $\delta_L(P)$ for the signed distance from P to a line L in that plane. Since

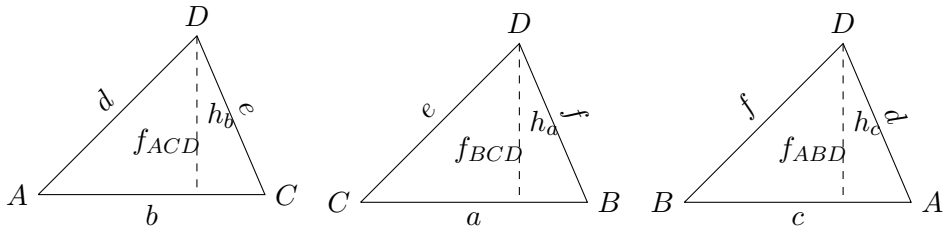


Figure 4: The three side faces of the tetrahedron.

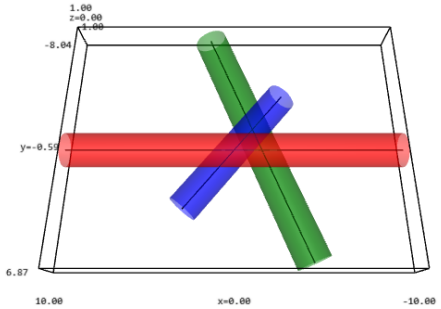


Figure 5: Three cylinders whose axes are the sides of the base triangle.

the base lies in the plane $z = 0$, Pythagoras gives

$$\operatorname{dist}(D, L)^2 = \delta_L(P)^2 + z^2.$$

Concretely, with $u^\perp = (-\sin\gamma, \cos\gamma)$ and $w^\perp$ a unit normal to AB,

$$\begin{aligned} C_a &: \ y^2 + z^2 = h_a^2, \\ C_b &: \ (u^\perp \cdot P)^2 + z^2 = h_b^2, \\ C_c &: \ \big(w^\perp \cdot (P - B)\big)^2 + z^2 = h_c^2. \end{aligned}$$

Subtracting the first equation from the other two eliminates z and leaves two conics in the plane:

$$\begin{aligned} Q_1 &: \ \delta_{CA}(P)^2 - \delta_{CB}(P)^2 = h_b^2 - h_a^2, \\ Q_2 &: \ \delta_{AB}(P)^2 - \delta_{CB}(P)^2 = h_c^2 - h_a^2. \end{aligned}$$

Each left-hand side factors as a product of two linear forms, for instance $\delta_{CA}^2 - \delta_{CB}^2 = (\delta_{CA} - \delta_{CB})(\delta_{CA} + \delta_{CB})$. The asymptotes (or, if the constant is 0, the components) of Q_1 are therefore parallel to the two angle bisectors at C. Likewise, those of Q_2 are parallel to the two angle bisectors at B.

Lemma 3.1. *Q_1 and Q_2 have no common component.*

Proof. A common component would be a common line, so some angle bisector at C would be parallel to some angle bisector at B. The internal bisectors at B and C meet at the incentre I, so they are not parallel. The external bisector at each vertex is perpendicular to the internal one there. So the only other possibility is that the internal bisectors at B and C are perpendicular, i.e. $\angle BIC = \pi/2$. But $\angle BIC = \frac{\pi}{2} + \frac{1}{2}\angle CAB > \frac{\pi}{2}$. □

Theorem 3.2. *Given the four face areas and two edges a, b meeting at a vertex, there are at most 4 tetrahedra up to isometry for each of the two possible base triangles, hence at most 8 in total. In particular, the volume takes at most 8 values.*

Proof. By Lemma 3.1 and Bézout's theorem for plane curves (Appendix A), the two conics Q_1, Q_2 meet in at most $2 \cdot 2 = 4$ points $P = (x, y)$. Conversely, any point D on all three cylinders satisfies Q_1 and Q_2 together with $z^2 = h_a^2 - y^2$. So each intersection point gives at most the two points $(x, y, \pm z)$, which are exchanged by the reflection $z \mapsto -z$. Up to isometry this leaves at most 4 tetrahedra for each base triangle, and there are at most 2 base triangles. The volume is $V = \frac{1}{3} f_{ABC}\,|z|$, so it takes at most 8 values. □

Remark 3.3. One can instead homogenise the three cylinder equations, e.g. $y^2 + z^2 - h_a^2 w^2 = 0$, to obtain three quadric surfaces in $\mathbb{P}^3(\mathbb{C})$. Bézout's theorem in $\mathbb{P}^3$ then bounds the number of intersection points by $2 \cdot 2 \cdot 2 = 8$, provided the intersection is finite, and counting complex points with multiplicity. The planar argument above is sharper: it proves finiteness, and it explains the bound 8 as 4 points in the plane times the mirror pair $\pm z$.

4 Volume of a d-simplex in terms of its $(d-2)$-faces

4.1 First observations

Remark 4.1 (Bounds). One might hope that V is bounded by the volumes of the regular simplices whose faces all equal $f_{\min} = \min_I f_I$ or $f_{\max} = \max_I f_I$. The lower bound is false. Take five points in $\mathbb{R}^3$ with no three collinear and regard them as a degenerate 4-simplex in $\mathbb{R}^4$. All ten triangles have positive area, so $f_{\min} > 0$, but the volume is 0. By continuity, nearby non-degenerate simplices have volume as small as we like while $f_{\min}$ stays bounded away from 0. We do not know whether the upper bound $V \le V_{\mathrm{reg}}(f_{\max})$ holds.

4.2 Regular d-simplex volume in terms of its $(d-2)$-faces

We use the standard formula for the regular n-simplex with edge length c:

$$V_n^2 = \frac{n+1}{(n!)^2\, 2^n}\, c^{2n}. \tag{R}$$

Theorem 4.2. *Let $d \ge 3$ and let R be a regular d-simplex whose $(d-2)$-faces all have volume f_c. Then*

$$V(R)^2 = \frac{d+1}{(d!)^2} \left(\frac{((d-2)!)^2}{d-1} \right)^{\frac{d}{d-2}} f_c^{\frac{2d}{d-2}}.$$

Proof. Let c be the common edge length. Each $(d-2)$-face is a regular $(d-2)$-simplex with edge c, so by (R) with $n = d-2$,

$$f_c^2 = \frac{d-1}{((d-2)!)^2\, 2^{d-2}}\, c^{2(d-2)}, \qquad \text{hence} \qquad c^2 = \left(\frac{((d-2)!)^2\, 2^{d-2}}{d-1}\, f_c^2 \right)^{\frac{1}{d-2}}.$$

Substituting into (R) with $n = d$ and writing $c^{2d} = (c^2)^d$:

$$V(R)^2 = \frac{d+1}{(d!)^2\, 2^d} \left(\frac{((d-2)!)^2\, 2^{d-2}}{d-1} \right)^{\frac{d}{d-2}} f_c^{\frac{2d}{d-2}}.$$

Since $(2^{d-2})^{d/(d-2)} = 2^d$, the powers of 2 cancel, giving the stated formula. □

For $d = 4$ this reads $V^2 = \frac{5}{324} f_c^4$. As a check, the regular 4-simplex with unit edge has $V^2 = \frac{5}{9216}$, and its faces have $f_c = \frac{\sqrt{3}}{4}$, so $\frac{5}{324} \cdot \frac{9}{256} = \frac{5}{9216}$.

4.3 Simplices with one distinguished edge

Definition 4.3. A d-simplex is *highly regular* if, for some labelling of its vertices, $l_{01} = a^2$ and $l_{ij} = b^2$ for every other edge. In other words, one edge has length a and all others have length b.

A $(d-2)$-face of a highly regular simplex either contains both p_0 and p_1 or it does not. In the first case it is itself highly regular, with volume f_a. In the second it is regular, with volume f_b. There are $\binom{d-1}{2}$ faces of the first kind. For $d = 4$ the face-set is

$$f_{012} = f_{013} = f_{014} = f_a, \qquad f_I = f_b \text{ for the other seven } I.$$

(A simplex with this face-set need not itself be highly regular.)

Lemma 4.4. *Let $n \geq 2$ and let $S_n(t,s)$ be the n-simplex with $l_{01} = t$ and $l_{ij} = s$ for all other edges. It exists if and only if $0 < t < \frac{2n}{n-1}s$, and then*

$$V\big(S_n(t,s)\big)^2 = \frac{t\, s^{n-2}\,\big(2ns - (n-1)t\big)}{(n!)^2\, 2^n}.$$

Proof. By scaling it suffices to take $s = 1$. Let S be a regular $(n-2)$-simplex with unit edge, spanned by $p_2, \dots, p_n$, with circumcentre O and circumradius ρ, where $\rho^2 = \frac{n-2}{2(n-1)}$. The points at distance 1 from every vertex of S are exactly the points $O + w$ with w in the 2-plane W orthogonal to $\mathrm{aff}(S)$ and $|w|^2 = r^2 := 1 - \rho^2 = \frac{n}{2(n-1)}$. So $p_0 = O + w_0$ and $p_1 = O + w_1$ lie on a circle of radius r in W, and $t = |w_0 - w_1|^2$ ranges over $(0, 4r^2] = (0, \frac{2n}{n-1}]$. At $t = 4r^2$ the points p_0, O, p_1 are collinear and the simplex is degenerate. This proves the existence statement.

For the volume, take the edge vectors from p_0, replacing $p_i - p_0$ for $i \geq 3$ by $p_i - p_2$, which does not change the determinant:

$$p_2 - p_0 = (p_2 - O) - w_0, \qquad p_i - p_2 \ (3 \leq i \leq n), \qquad p_1 - p_0 = w_1 - w_0.$$

The vectors $p_i - p_2$ and $p_2 - O$ lie in the direction space U of $\mathrm{aff}(S)$, and $w_0, w_1 \in W = U^{\perp}$. The matrix of these n vectors is therefore block triangular with respect to $U \oplus W$. Hence

$$\begin{aligned} n!\, V &= \big|\det{}_U(p_3 - p_2, \dots, p_n - p_2)\big| \cdot \big|\det{}_W(-w_0,\, w_1 - w_0)\big| \\ &= (n-2)!\ \mathrm{vol}(S) \cdot 2\,\mathrm{area}(O, p_0, p_1). \end{aligned}$$

The triangle Op_0p_1 is isosceles with legs r and base $\sqrt{t}$, so area$^2 = \frac{t}{4}\big(r^2 - \frac{t}{4}\big) = \frac{t(2n-(n-1)t)}{16(n-1)}$. By (R), $\mathrm{vol}(S)^2 = \frac{n-1}{((n-2)!)^2 2^{n-2}}$. Combining,

$$V^2 = \frac{4((n-2)!)^2}{(n!)^2}\cdot\frac{n-1}{((n-2)!)^2 2^{n-2}}\cdot\frac{t(2n-(n-1)t)}{16(n-1)} = \frac{t\,(2n-(n-1)t)}{(n!)^2\,2^n}. \quad \square$$

For $n = 2$ this is the familiar area$^2 = \frac{a^2(4b^2-a^2)}{16}$ for an isosceles triangle.

Theorem 4.5. *Let $d \geq 4$ and let T be highly regular with face values f_a, f_b. Put $\rho = f_a/f_b$, $K = ((d-2)!)^2\,2^{d-2}$ and*

$$\Delta = (d-2)^2 - (d-1)(d-3)\rho^2.$$

Then $b^2 = \left(\frac{Kf_b^2}{d-1}\right)^{1/(d-2)}$, and $a^2 = \tau\, b^2$ where τ is one of the two roots

$$\tau_\pm = \frac{(d-2) \pm \sqrt{\Delta}}{d-3}.$$

Accordingly,

$$V(T)^2 = \frac{\tau\,(2d-(d-1)\tau)}{(d!)^2\,2^d}\left(\frac{Kf_b^2}{d-1}\right)^{\frac{d}{d-2}}, \qquad \tau \in \{\tau_+, \tau_-\}.$$

Proof. The faces with value f_b are regular with edge b, so $f_b^2 = (d-1)\,b^{2(d-2)}/K$ by (R), which gives b^2. The faces with value f_a are copies of $S_{d-2}(a^2, b^2)$, so by Lemma 4.4, with $s = b^2$ and $\tau = a^2/b^2$,

$$Kf_a^2 = s^{d-2}\,\tau\big(2(d-2) - (d-3)\tau\big).$$

Dividing by $Kf_b^2 = (d-1)s^{d-2}$ gives the quadratic

$$(d-3)\tau^2 - 2(d-2)\tau + (d-1)\rho^2 = 0,$$

whose roots are $\tau_\pm$. Finally, Lemma 4.4 with $n = d$ gives $V^2 = s^d\,\tau(2d - (d-1)\tau)/((d!)^2 2^d)$ with $s^d = \big(Kf_b^2/(d-1)\big)^{d/(d-2)}$. $\square$

The two roots satisfy $\tau_+ + \tau_- = \frac{2(d-2)}{d-3}$, so they are symmetric about

$$\tau_0 = \frac{d-2}{d-3}.$$

Each root gives a genuine simplex exactly when $0 < \tau < \frac{2d}{d-1}$, by Lemma 4.4. The regular simplex is the case $\rho = 1$, $\tau_- = 1$, where Theorem 4.5 recovers Theorem 4.2.

4.3.1 The case $d = 4$

Here $K = 16$, $\tau_\pm = 2 \pm \sqrt{4 - 3\rho^2}$ and $(K f_b^2/3)^2/((4!)^2 2^4) = f_b^4/324$. Writing $u = \sqrt{4 - 3\rho^2}$, we get $\tau_\mp(8 - 3\tau_\mp) = (2 \mp u)(2 \pm 3u) = 9\rho^2 - 8 \pm 4u$, and therefore

$$V^2 = \frac{9 f_a^2 f_b^2 - 8 f_b^4 \pm 4 f_b^3 \sqrt{4 f_b^2 - 3 f_a^2}}{324}. \tag{1}$$

The upper sign corresponds to the shorter edge, $a^2 = \tau_- b^2$, and is realised for every $0 < \rho^2 \leq \frac{4}{3}$. The lower sign corresponds to $a^2 = \tau_+ b^2$, and is realised exactly when $\tau_+ < \frac{8}{3}$, i.e. when

$$\frac{32}{27} < \frac{f_a^2}{f_b^2} < \frac{4}{3}.$$

In that range *both* signs occur, and they give different volumes, since the two values differ by $\frac{8}{324} f_b^3 \sqrt{4 f_b^2 - 3 f_a^2} \neq 0$. Squaring out the $\pm$ gives the single polynomial relation

$$\left(324 V^2 - 9 f_a^2 f_b^2 + 8 f_b^4\right)^2 = 16 f_b^6 \left(4 f_b^2 - 3 f_a^2\right),$$

which is quadratic in V^2, in line with Corollary 2.9.

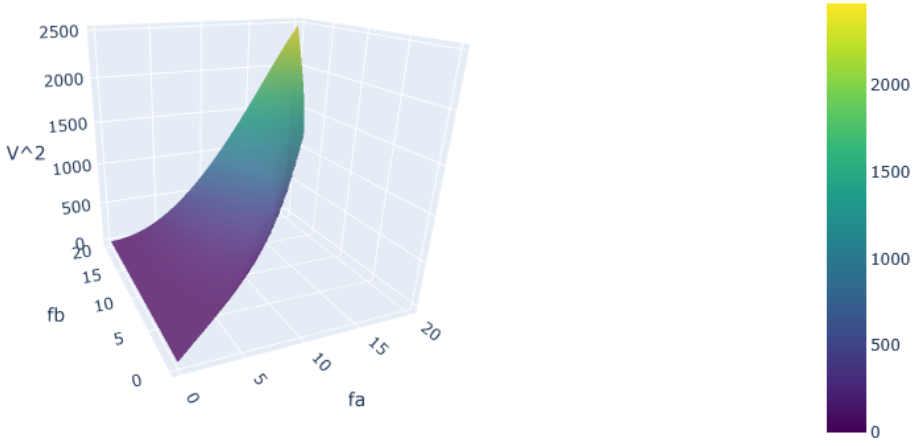


Figure 6: Plot of the upper-sign branch of (1).

4.4 The face-set does not determine the volume

The two branches in Theorem 4.5 answer Question 2.5 negatively. This construction is due to Mohar and Rivin [1], who observed that the face volumes of this family are symmetric about τ_0. We give a self-contained statement and proof.

Theorem 4.6. *Let $d \geq 4$, let $t_0 = \frac{d-2}{d-3}$, and let*

$$t \in \left(\frac{4}{(d-1)(d-3)},\ \frac{2d}{d-1} \right), \qquad t \neq t_0.$$

Let $T_t = S_d(t, 1)$ be the d-simplex with $l_{01} = t$ and all other $l_{ij} = 1$, and let $t' = 2t_0 - t$. Then T_t and $T_{t'}$ are both non-degenerate and have the same labelled face-set, $f_I(T_t) = f_I(T_{t'})$ for every I. However,

$$V(T_t)^2 - V(T_{t'})^2 = \frac{8\,(t_0 - t)}{(d-3)\,(d!)^2\,2^d} \neq 0.$$

In particular, $T_t \ncong T_{t'}$, and V is not a function of the face-set.

Proof. Existence. By Lemma 4.4, we need $0 < t, t' < \frac{2d}{d-1}$. For t this is given. For t': the condition $t' > 0$ means $t < 2t_0$, which holds because $\frac{2d}{d-1} < \frac{2(d-2)}{d-3}$, i.e. $d^2 - 3d < d^2 - 3d + 2$. The condition $t' < \frac{2d}{d-1}$ means

$$t > 2t_0 - \frac{2d}{d-1} = \frac{2(d-2)(d-1) - 2d(d-3)}{(d-3)(d-1)} = \frac{4}{(d-1)(d-3)}.$$

The interval is non-empty and contains t_0 because $d \geq 4$.

Equal faces. A $(d-2)$-face not containing both p_0, p_1 has all edges equal to 1 in both simplices. A face containing both is $S_{d-2}(t, 1)$, resp. $S_{d-2}(t', 1)$. By Lemma 4.4 with $n = d - 2$,

$$f^2 = \frac{t\,\big(2(d-2) - (d-3)t\big)}{((d-2)!)^2\,2^{d-2}}.$$

This quadratic in t has roots 0 and $2t_0$, so it is symmetric under $t \mapsto 2t_0 - t$.

Different volumes. By Lemma 4.4 with $n = d$, $(d!)^2 2^d\,V^2 = g(t) := 2dt - (d-1)t^2$. Then

$$\begin{aligned} g(t) - g(t') &= (t - t')\big(2d - (d-1)(t + t')\big) = (t - t') \left(2d - \frac{2(d-1)(d-2)}{d-3} \right) \\ &= -\frac{4(t - t')}{d-3}, \end{aligned}$$

and $t - t' = 2(t - t_0)$. □

Example 4.7 ($d = 4$)**.** Here $t_0 = 2$ and the admissible range is $\frac{4}{3} < t < \frac{8}{3}$. Take $t = \frac{3}{2}$, so $t' = \frac{5}{2}$:

- $T_{3/2}$: one edge of length $\sqrt{3/2}$, nine of length 1;
- $T_{5/2}$: one edge of length $\sqrt{5/2}$, nine of length 1.

In both, the three triangles $012, 013, 014$ have $f^2 = \frac{15}{64}$ and the other seven have $f^2 = \frac{3}{16}$. But

$$V(T_{3/2})^2 = \frac{21}{36864} \approx 5.70 \times 10^{-4}, \qquad V(T_{5/2})^2 = \frac{5}{36864} \approx 1.36 \times 10^{-4}.$$

These values agree with (1) (upper and lower signs respectively), and we also checked them numerically with the Cayley–Menger determinant.

4.5 Local rigidity and the Jacobian

Since each f_I^2 is homogeneous of degree $d-2$ in the squared lengths $l = (l_{ij})$, Euler's theorem relates Φ to its Jacobian

$$J(l) = \left(\frac{\partial f_I^2}{\partial l_{ij}} \right)_{I,\{i,j\}},$$

a $k \times k$ matrix.

Theorem 4.8 (Euler's homogeneous function theorem)**.** *If g is continuously differentiable on an open cone in $\mathbb{R}^n$ and positively homogeneous of degree h, then $\sum_{i=1}^{n} x_i \frac{\partial g}{\partial x_i} = h\, g$.*

Applying this to each f_I^2 gives

$$J(l)\, l = (d-2)\, \Phi(l).$$

(If one uses edge lengths instead of squared lengths, the degree is $2(d-2)$.) Some consequences:

- By Theorem 2.8, $\det J$ is a non-zero polynomial in l. So $J(l)$ is invertible for l off the proper algebraic subset $Z = \{\det J = 0\}$, the *critical locus*. Note that $\det J$ is not non-zero everywhere.
- Off Z, $l = (d-2)\, J(l)^{-1} \Phi(l)$. This is an implicit identity, since J itself depends on l, and not a change of variables. It does not by itself give a formula for V in terms of F.

- By the inverse function theorem, Φ is a local diffeomorphism near every $l \notin Z$. Since l determines T up to congruence, every simplex off Z is *locally rigid*: any simplex with the same face-set that is sufficiently close to it is congruent to it. In particular, the regular simplex is locally rigid (Theorem 2.8).

This shows that Theorem 4.6 is a genuinely *discrete* phenomenon: T_t and $T_{t'}$ are far apart, and each is isolated among simplices with its face-set (for generic t).

Question 4.9 (Continuous version; see Zhang [4])**.** Let $d \geq 4$. If a continuous deformation of a d-simplex in $\mathbb{R}^d$ preserves the volumes of all its $(d-2)$-faces, must it be a rigid motion?

By the local rigidity above, a non-trivial deformation $l(s)$ is constant on every interval of parameters s with $l(s) \notin Z$. So all of its trace, apart from at most countably many points, lies in the critical locus Z. The question is therefore concentrated on the structure of Z. Zhang [4] shows that the answer is *no* in suitable pseudo-Euclidean spaces $\mathbb{R}^{p,d-p}$. For example, for $d = 4$ in $\mathbb{R}^{3,1}$ there is a continuous family of non-congruent 4-simplices with fixed 2-face areas. In Euclidean space the question remains open.

5 Conclusion

1. **The discrete question has a negative answer.** The $(d-2)$-face volumes do not determine the volume of a d-simplex, even when the faces are labelled. For every $d \geq 4$, Theorem 4.6 gives explicit pairs T_t, T_{2t_0-t} with identical face-sets and different volumes; for $d = 4$, Example 4.7 gives the pair with one edge of length $\sqrt{3/2}$, resp. $\sqrt{5/2}$, and all other edges of length 1. The construction arises naturally from the two roots of the highly regular formula, and is the family already given by Mohar and Rivin [1]. In particular, there is no formula $V^2 = p(F)$, polynomial or otherwise.

2. **Generic finiteness.** By the algebraic independence of the f_I^2 (Mohar–Rivin) and a dimension count, V^2 satisfies a polynomial relation $q(F^2, V^2) = 0$ of positive degree in V^2. Hence, generically, only finitely many volumes share a face-set (Corollary 2.9). The right object is therefore a *multivalued* algebraic function, as studied by Gaifullin [3]. In the highly regular family for $d = 4$ this function is explicitly two-valued, and satisfies a relation quadratic in V^2.

3. **Special formulas.** For regular simplices (Theorem 4.2) and highly regular simplices (Theorem 4.5) we obtained closed forms for V in terms of the face volumes.

4. **Tetrahedra.** The four face areas and two edges meeting at a vertex determine a tetrahedron up to at most 8 possibilities, at most 4 for each of the two possible base triangles (Theorem 3.2).

5. **Contrast with edges.** The edge lengths determine a simplex uniquely up to congruence, while the equally numerous $(d-2)$-face volumes determine it only up to finitely many choices, generically.

6. **The continuous question is open.** Every simplex off the critical locus $Z = \{\det J = 0\}$, including the regular simplex, is locally rigid. Whether a non-trivial continuous deformation preserving all $(d-2)$-face volumes can exist in Euclidean $\mathbb{R}^d$ (Question 4.9) remains open; Zhang [4] constructs such deformations in pseudo-Euclidean signatures. Any Euclidean counterexample must essentially live inside Z. This is the natural next direction, and the subject of our continuing work.

A Bézout's theorem

Plane version. If two plane curves of degrees m and n (over $\mathbb{R}$ or $\mathbb{C}$) have no common component, they meet in at most mn points [5]. This is the version used in Theorem 3.2.

Projective version. Let n hypersurfaces in $\mathbb{P}^n$ over an algebraically closed field be defined by homogeneous polynomials of degrees $l_1, \ldots, l_n$. Then either they have infinitely many common points, or the number of common points, counted with multiplicity, equals $l_1 \cdots l_n$. In particular, the number of distinct common points is then at most $l_1 \cdots l_n$.

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[5] W. Fulton, *Algebraic Curves: An Introduction to Algebraic Geometry*, Benjamin, 1969.

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